

Discriminant Analysis

The DISCRIM Procedure

Total Sample Size	572	DF Total	571
Variables	8	DF Within Classes	569
Classes	3	DF Between Classes	2

Number of Observations Read	572
Number of Observations Used	572

Class Level Information					
region	Variable Name	Frequency	Weight	Proportion	Prior Probability
1	1	323	323.0000	0.564685	0.333333
2	2	98	98.0000	0.171329	0.333333
3	3	151	151.0000	0.263986	0.333333

Within Covariance Matrix Information		
region	Covariance Matrix Rank	Natural Log of the Determinant of the Covariance Matrix
1	8	51.57703
2	8	40.30530
3	8	47.66458
Pooled	8	52.43120

Discriminant Analysis

The DISCRIM Procedure Test of Homogeneity of Within Covariance Matrices

Chi-Square	DF	Pr > ChiSq
2110.747587	72	<.0001

Since the Chi-Square value is significant at the 0.1 level, the within covariance matrices will be used in the discriminant function.
Reference: Morrison, D.F. (1976) Multivariate Statistical Methods p252.

Discriminant Analysis

The DISCRIM Procedure

Generalized Squared Distance to region			
From region	1	2	3
1	51.57703	1269	1591
2	89.09331	40.30530	81.29521
3	76.08996	96.69547	47.66458

Discriminant Analysis

The DISCRIM Procedure Classification Summary for Calibration Data: WORK.OIL Resubstitution Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into region

From region	1	2	3	Total
1	323 100.00	0 0.00	0 0.00	323 100.00
2	0 0.00	98 100.00	0 0.00	98 100.00
3	0 0.00	0 0.00	151 100.00	151 100.00
Total	323 56.47	98 17.13	151 26.40	572 100.00
Priors	0.33333	0.33333	0.33333	

Error Count Estimates for region				
	1	2	3	Total
Rate	0.0000	0.0000	0.0000	0.0000
Priors	0.3333	0.3333	0.3333	

Discriminant Analysis

The DISCRIM Procedure
 Classification Summary for Calibration Data: WORK.OIL
 Cross-validation Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into region				
From region	1	2	3	Total
1	323 100.00	0 0.00	0 0.00	323 100.00
2	0 0.00	98 100.00	0 0.00	98 100.00
3	0 0.00	0 0.00	151 100.00	151 100.00
Total	323 56.47	98 17.13	151 26.40	572 100.00
Priors	0.33333	0.33333	0.33333	

Error Count Estimates for region				
	1	2	3	Total
Rate	0.0000	0.0000	0.0000	0.0000
Priors	0.3333	0.3333	0.3333	

Discriminant analysis — nine-area classification

The DISCRIM Procedure

Total Sample Size	572	DF Total	571
Variables	8	DF Within Classes	563
Classes	9	DF Between Classes	8

Number of Observations Read	572
Number of Observations Used	572

Class Level Information

area	Variable Name	Frequency	Weight	Proportion	Prior Probability
Calabria	Calabria	56	56.0000	0.097902	0.111111
Coast-Sardin	Coast-Sardin	33	33.0000	0.057692	0.111111
East-Liguria	East-Liguria	50	50.0000	0.087413	0.111111
Inland-Sardi	Inland-Sardi	65	65.0000	0.113636	0.111111
North-Apulia	North-Apulia	25	25.0000	0.043706	0.111111
Sicily	Sicily	36	36.0000	0.062937	0.111111
South-Apulia	South-Apulia	206	206.0000	0.360140	0.111111
Umbria	Umbria	51	51.0000	0.089161	0.111111
West-Liguria	West-Liguria	50	50.0000	0.087413	0.111111

Within Covariance Matrix Information		
area	Covariance Matrix Rank	Natural Log of the Determinant of the Covariance Matrix
Calabria	8	45.53284
Coast-Sardin	8	34.57630
East-Liguria	8	42.87014
Inland-Sardi	8	37.97195
North-Apulia	8	45.19879
Sicily	8	48.68371
South-Apulia	8	48.35201
Umbria	8	31.98920
West-Liguria	8	43.43016
Pooled	8	48.77624

Discriminant analysis — nine-area classification

The DISCRIM Procedure
Test of Homogeneity of Within Covariance Matrices

Chi-Square	DF	Pr > ChiSq
2689.722277	288	<.0001

Since the Chi-Square value is significant at the 0.1 level, the within covariance matrices will be used in the discriminant function.
Reference: Morrison, D.F. (1976) Multivariate Statistical Methods p252.

Discriminant analysis — nine-area classification

The DISCRIM Procedure

Generalized Squared Distance to area									
From area	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria
Calabria	45.53284	1986	2065	1534	102.06921	63.03648	71.16537	3930	1977
Coast-Sardin	204.81354	34.57630	197.34800	110.01413	253.37632	171.05501	93.34046	4356	219.09214
East-Liguria	111.43362	477.26843	42.87014	263.78791	99.82658	108.54161	102.88254	216.77091	116.78191
Inland-Sardi	163.93381	83.49209	125.35352	37.97195	147.79400	120.85087	85.96622	2195	188.63474
North-Apulia	108.09092	3183	2942	2289	45.19879	55.87050	83.70884	2515	2817
Sicily	59.43233	3577	3725	2760	86.94219	48.68371	69.56056	4892	3505
South-Apulia	94.80073	1419	2271	1138	167.51953	82.72006	48.35201	8063	1500

Generalized Squared Distance to area									
From area	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria
Umbria	98.87493	550.31284	57.26577	401.97420	114.44790	106.69007	105.29592	31.98920	118.88048
West-Liguria	172.49356	232.48480	84.02640	190.98996	150.53452	271.75506	105.67240	922.86643	43.43016

Discriminant analysis — nine-area classification

The DISCRIM Procedure
 Classification Summary for Calibration Data: WORK.OIL
 Resubstitution Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into area										
From area	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria	Total
Calabria	54 96.43	0 0.00	0 0.00	0 0.00	0 0.00	1 1.79	1 1.79	0 0.00	0 0.00	56 100.00
Coast-Sardin	0 0.00	33 100.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	33 100.00
East-Liguria	0 0.00	0 0.00	50 100.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	50 100.00
Inland-Sardi	0 0.00	0 0.00	0 0.00	65 100.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	65 100.00
North-Apulia	0 0.00	0 0.00	0 0.00	0 0.00	25 100.00	0 0.00	0 0.00	0 0.00	0 0.00	25 100.00
Sicily	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	36 100.00	0 0.00	0 0.00	0 0.00	36 100.00
South-Apulia	1 0.49	0 0.00	0 0.00	0 0.00	0 0.00	5 2.43	200 97.09	0 0.00	0 0.00	206 100.00
Umbria	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	51 100.00	0 0.00	51 100.00
West-Liguria	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	50 100.00	50 100.00
Total	55 9.62	33 5.77	50 8.74	65 11.36	25 4.37	42 7.34	201 35.14	51 8.92	50 8.74	572 100.00
Priors	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	

Error Count Estimates for area										
	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria	Total
Rate	0.0357	0.0000	0.0000	0.0000	0.0000	0.0000	0.0291	0.0000	0.0000	0.0072
Priors	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	

Discriminant analysis — nine-area classification

The DISCRIM Procedure
 Classification Summary for Calibration Data: WORK.OIL
 Cross-validation Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into area										
From area	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria	Total
Calabria	53 94.64	0 0.00	0 0.00	0 0.00	0 0.00	2 3.57	1 1.79	0 0.00	0 0.00	56 100.00
Coast-Sardin	0 0.00	32 96.97	0 0.00	0 0.00	0 0.00	0 0.00	1 3.03	0 0.00	0 0.00	33 100.00

Number of Observations and Percent Classified into area										
From area	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria	Total
East-Liguria	0 0.00	0 0.00	49 98.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	1 2.00	50 100.00
Inland-Sardi	0 0.00	1 1.54	0 0.00	64 98.46	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	65 100.00
North-Apulia	0 0.00	0 0.00	0 0.00	0 0.00	23 92.00	1 4.00	1 4.00	0 0.00	0 0.00	25 100.00
Sicily	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	34 94.44	2 5.56	0 0.00	0 0.00	36 100.00
South-Apulia	1 0.49	0 0.00	0 0.00	0 0.00	0 0.00	5 2.43	200 97.09	0 0.00	0 0.00	206 100.00
Umbria	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	51 100.00	0 0.00	51 100.00
West-Liguria	0 0.00	0 0.00	1 2.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	49 98.00	50 100.00
Total	54 9.44	33 5.77	50 8.74	64 11.19	23 4.02	42 7.34	205 35.84	51 8.92	50 8.74	572 100.00
Priors	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	0.11111	

Error Count Estimates for area										
	Calabria	Coast-Sardin	East-Liguria	Inland-Sardi	North-Apulia	Sicily	South-Apulia	Umbria	West-Liguria	Total
Rate	0.0536	0.0303	0.0200	0.0154	0.0800	0.0556	0.0291	0.0000	0.0200	0.0338
Priors	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	